## 3.1.8 Passing arguments by value and by reference

PHP supports two ways of passing arguments to functions. The default way is passing arguments by value. When we pass arguments by value, the function works only with the copies of the values. This may lead to performance overheads when we work with large amounts of data.

When we pass values by reference, the function receives a reference to the actual value. The original values are affected when modified. This way of passing values is more time and space efficient. On the other hand, it is more error prone.

Which way of passing arguments should we use? It depends on the situation. Say we have a set of data, salaries of employees. If we want to compute some statistics of the data, we do not need to modify them. We pass by values. If we work with large amounts of data and the speed of computation is critical, we pass by reference. If we want to modify the data, e.g. do some reductions or raises to the salaries, we might pass by reference.

The following two examples cover both concepts.

swap1.php

```
<?php

function swap($a, $b) {

    $temp = $a;
    $a = $b;
    $b = $temp;
    echo "inside swap function:\n";
    echo "\$a is $a \n";
    echo "\$b is $b \n";
}

$a = 4;
$b = 7;
```

**Did You Know?**

In 1993, the National Center for Supercomputing Applications (NCSA), a unit of the University of Illinois at Urbana-Champaign, released NCSA Mosaic, the first popular graphical Web browser, which played an important part in expanding the growth of the nascent World Wide Web.